LAB WORKSHEET

Cryptography Lab

How cryptography protects data: ciphers, hashing, password storage, encryption, signatures, certificates and HTTPS.

Key ideas

- **Caesar** = fixed shift, only 25 keys. Break it by brute force or **frequency analysis** (E, T, A most common). Vigenère uses a repeating keyword.
- **Hashing (SHA-256)**: one-way, fixed length, collision-resistant. The avalanche effect: one changed character flips about half the bits. Compare hashes to prove a file is unchanged. SHA-1 is broken.
- **Passwords**: never store plaintext. Use a unique **salt** per user and a **slow hash** (PBKDF2, bcrypt, Argon2). Salts aren't secret; they defeat precomputed tables.
- **Symmetric (AES-GCM)**: one shared key plus a random IV; the tag detects tampering. The problem is sharing the key; Diffie–Hellman solves it.
- **Asymmetric**: encrypt with the recipient's **public** key, decrypt with their **private** key. Sign with your private key; anyone verifies with your public key (integrity, authenticity, non-repudiation).
- **HTTPS**: check the name/SAN, validity dates and the chain to a trusted root. In TLS 1.3, after ClientHello/ServerHello with key shares, everything is encrypted.

Guided steps

Encrypt with Caesar → break a message by frequency → avalanche effect → integrity check → salts → PBKDF2 → AES round trip and wrong key → RSA → tampered signature → certificate diagnosis → TLS handshake.

Independent task

Secure messaging investigation for Kestrel Moor Logistics (fictional): eight challenges using the lab's own tools.

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